

ECE498A: CAPSTONE DESIGN PROJECT

S2026

COURSE OUTLINE*

Department of Electrical and Computer Engineering
University of Waterloo

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The teaching team gratefully acknowledges the help of Bill Bishop and Dan Davison in organizing this course. Much of the content of this Course Outline is taken, adopted, or inspired from their work.

COURSE OVERVIEW

ECE498A is the first course in the two-course sequence ECE498A -- ECE498B that focuses on a major ECE design project.

Course Outcomes

By the end of ECE498B, ideally you should be able to:

- define an engineering design problem in terms of an objective and design specifications
- make, and clearly justify, substantial design decisions
- apply advanced engineering knowledge to a new engineering design problem
- use engineering tools to construct and debug hardware and/or software
- manage time and resources to meet tight deadlines
- function effectively in a team, appreciate how group dynamics evolve over time, and prevent or resolve inter-personal conflict
- evaluate risks and devise contingency plans.

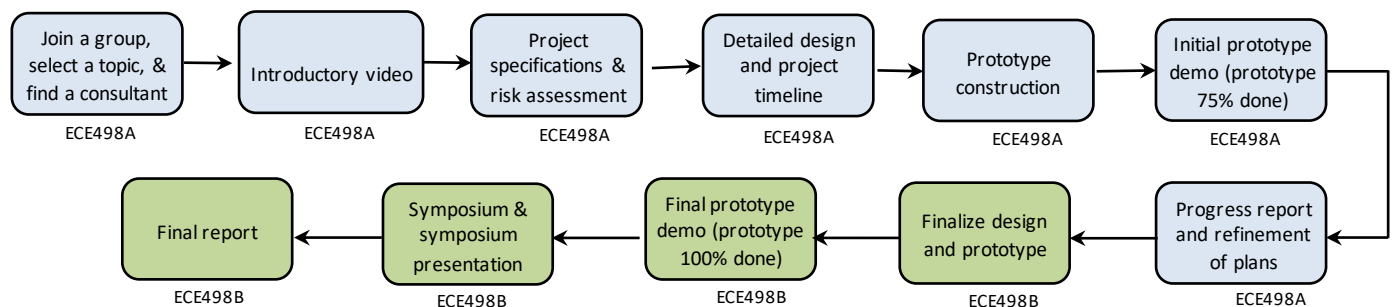
Guiding Principles

There are five guiding principles that you should keep in mind when choosing your project and when submitting deliverables:

- the project must involve engineering design (i.e., a creative, iterative, and open-ended process)
- it must involve quantitative technical analysis (i.e., quantitative investigation using engineering theory, simulations, and/or experimental data)
- **It must involve at least 2 of the core electrical/computer engineering components in a substantial way: communications, digital circuits, electronic devices, circuits, programming, power systems, RF, controls.**
- it must be at an advanced level (i.e., require upper-year engineering knowledge for design and/or analysis and be complicated enough to require, over the two terms, about 960 hours of work for 4-group teams, or 1200--1440 hours of work for 5/6-group teams).
- The project must have a hardware component in the final prototype.

Milestones

Main milestones in the courses are shown in the diagram below:



Details about each milestone are explained in this document and in the Course Outline for ECE498B.

CONTACT INFORMATION

Name	Title and Responsibilities	Contact Information
Otman Basir Jeff Zarnett Ayman El-Hag	Course Instructors - course deliverables, submission deadlines, grades	otman.basir@uwaterloo.ca jeff.zarnett@uwaterloo.ca ahalhaj@uwaterloo.ca
Ayman El-Hag	Course Coordinator - course logistics, enrollment, connections to ECE498B	ahalhaj@uwaterloo.ca EIT-4016

The best way to contact them is by email. Please indicate your group number (e.g., Group 2024.75) in all correspondence.

ECE PROJECT ROOM (E2-3353)

E2-3353 is the ECE project lab room, open to all ECE students in 2A--4B, intended as a resource primarily for undergraduate students working on open-ended projects. Please pay attention to the following rules and protocols:

- **Safety:** See the important document called *Safety Rules for E2-3353* posted on LEARN and in the project room. These rules will be reviewed in class. The rules are updated occasionally; the latest version will always be posted on LEARN and in the project room.
- **Protocol for sharing the room:**
 1. If a class is scheduled to use the room for a lab, students enrolled in the class have the highest priority. During those hours, other undergraduate ECE students may use the room for project work, but only if benches are available and if the students work quietly.
 2. During all other hours, when the room is open and there is no scheduled lab, ECE students doing project work for ECE498A/B have the highest priority. The next priority is for undergraduate ECE students working on extra-curricular projects. Students doing course work or purely computer work are not permitted to use the room unless benches are available.
 3. If the room is under high demand, please share the space and resources respectfully.
- **Rules to keep the room clean and secure:**
 1. If you were given a code to access the room, do not share the code with anyone else.
 2. Do not prop the door open unless a lab staff member permits it.
 3. Keep the windows closed to prevent flooding from rain (in 4A) or freezing pipes (in 4B).
 4. Leave equipment and furniture in place.
 5. Keep all food and drink outside the room.
 6. Keep the space clean and tidy.
 7. The department reserves the right to restrict access to the room for safety or security reasons.
- **Storage:** There are shelves in E2-3353 that you may use to store prototype parts, but because the space is not locked, we recommend all valuable or delicate parts be kept in locked storage (e.g., the small lockers outside of the lab), or that you store them off campus. Space in E2-3353 is limited, and if it fills up, the Lab Instructor has authority to decide how space is allocated. The ECE department is not responsible for any damage or theft of prototype parts kept in storage in E2-3353 or in the nearby lockers.
- **Soldering:** Prior to using a soldering station, you are required to review an on-line tutorial on effective and safe soldering. See the document *Safety Rules for Project Room E2-3353* for rules that must be followed when soldering.
- **Drilling, sanding, filing, painting:** Use the student design shop in E5-1101D for activities such as drilling, sanding, filing or painting materials. You may not do these activities in the ECE project lab room. You must follow the student shop safety protocols (see <https://uwaterloo.ca/engineering-machine-shop/>).

SAFETY INFORMATION

University Safety Office

The University's Safety Office has some good safety resources, including training sessions (see <http://www.safetyoffice.uwaterloo.ca/>). One weakness is that they don't have training for electrical or mechanical safety.

Safety: General Principles, Electrical Safety, Mechanical Safety

Here are some key points that everyone should keep in mind:

A. University employees and students are required to work in a safe manner and to encourage others to work in a safe manner. Why?

1. According to the Criminal Code of Canada "Everyone is criminally negligent who in doing anything, or in omitting to do anything that it is his duty to do, shows wanton or reckless disregard for the lives or safety of other persons" 219.(1).
2. Engineers are held to a higher safety standard since they are expected to be familiar with safety and risk mitigation principles. Failure to act to correct a potentially dangerous situation, and failure to follow codes and standards, are examples of professional misconduct for engineers (see R.R.O. 1990, Reg. 941, Section 72 [Professional Misconduct \(Section 72 of the Regulation - O. Reg. 941\) | Professional Engineers Ontario \(peo.on.ca\)](#)).
3. There is Canadian Safety Legislation that affects everyone:
 - a. Canada Occupational Health and Safety Regulations (SOR/86-304) (see <http://laws.justice.gc.ca/eng/regulations/SOR-86-304/>)
 - b. Ontario Occupational Health and Safety Act (see <http://www.labour.gov.on.ca/english/hs/laws/index.php>)
4. Policy 34 of the university (see <https://uwaterloo.ca/secretariat/policies-procedures-guidelines/policy-34>) explicitly requires all employees and students to "comply with applicable legislation, University policies, guidelines and procedures, and additional directions from the University or a University employee in a position of authority, with respect to health and safety." The policy also states that each person at the university is responsible for their own safety and actions, and for the impact of those actions on others. Policy 60 (see <https://uwaterloo.ca/secretariat/policies-procedures-guidelines/policy-60>) has related emergency response information.

B. It's impossible to make a workplace 100% free of hazards. A key goal of engineering is to reduce the risk of injury or illness to an acceptable level.

Consequently, you must never be casual about safety issues. You need to think about safety as your work in E2-3353, as you construct your prototype, as you test it out, and as you demonstrate it to others.

C. Electrical hazards are especially relevant for most of your projects:

1. Do not overload an electrical system—overloading produces heat or arcing. Use proper gauge wiring. Install a fuse to limit current. Avoid metal jewelry, pens, and rulers.
2. Don't underestimate how dangerous a shock hazard can be. As little as 10 mA flowing through a muscle will cause that muscle to contract—you literally cannot let go of the wire! 100 mA can be fatal.

3. If your project includes

- a. custom 120V AC circuitry, or
- b. any other circuits (e.g., motor controllers) running at voltages above 40 V, or
- c. any circuit running at currents greater than 3 A, or
- d. any size of lead-acid battery

then before starting to build your circuit you must talk to one of the course instructors about incorporating safety into circuit design.

4. If your project satisfies either a., b., c., or d. in item 3 above, then you must work in pairs whenever you are probing or modifying the circuitry.
5. You need to ensure any circuits you design and build are safe. Be sure that:
- a. Circuits are enclosed in a plastic or metal box that is approved for electrical use.
 - b. Circuits have appropriate power switches and fuses.
 - c. High-capacity (e.g., lithium-ion or lead-acid) batteries have well-insulated connections, overload protection, and a means of disconnection.
 - d. Proper wires and connectors are used throughout.
 - e. Power cords entering electrical boxes have an approved means of strain relief.
 - f. All 120V AC devices, including power supplies/adapters, are approved by CSA, UL, ULC, or ESA.*
 - g. Circuits are appropriately grounded.

The Lab Instructor can provide more information on these aspects of circuit construction.

6. See the document *Safety Rules for E2-3353* on LEARN for more electrical safety information.

D. Mechanical hazards are relevant for some of your projects:

1. You need to build appropriate mechanical shielding into your prototype if your project involves:
 - a. moving parts that may pinch, hit, or crush a person, or
 - b. sharp edges or points, or
 - c. high-pressure gases or liquids.
2. You need to incorporate appropriate bracing into any part or structure that risks falling over.

Safety Rules for E2-3353

In addition to being aware of general safety information, in the ECE project room (E2-3353) there are some specific safety protocols that must be followed. See the document *Safety Rules for E2-3353* on LEARN. These rules will be reviewed in class. Pay attention and follow the rules to minimize the chance of injuries.

Remember: Everyone is responsible for safety.

Safety at the Design Symposium in ECE498B

Safety at the design symposium, held in March next year, is doubly important since the symposium is open to the public. Symposium safety is discussed in ECE498B, but you should start thinking right now about safety as you develop your project idea and build your prototype. Note that your prototype will be inspected for safety in ECE498B. Prototypes deemed to be unsafe will not be allowed to operate at the symposium.

***Warning:** Many power adapters purchased legally in Canada have only CE approval, which is not recognized in Canada. So do not buy a power adapter unless you can verify the certification status first. We recommend using Digikey Canada since you can verify on-line the certification status of virtually every item that they sell.

COURSE ACTIVITIES

Date	Time	Location	Topic/Event
May 12 (Tuesday)	4:00- 5:20pm	M3-1006	- Overview of the course, group work, project awards - Info on forming groups, selecting topics, finding consultants, keeping logs - Expectations for the Project Abstract and Project Agreement Form
May 19 (Tuesday)	4:00- 5:20pm	M3-1006	- Session for students who are not yet part of a complete group (a group is complete when it has 4--6 members)
May 26 (Tuesday)	4:00- 5:20pm	M3-1006	- Expectations for the Project Specifications & Risk Assessment document - Information on safety and the Safety Quiz
June 09 (Tuesday)	4:00- 5:20pm	M3-1006	- Expectations for the Detailed Design and Project Timeline Report
June 23 (Tuesday)	4:00- 5:20pm	M3-1006	- Tips for prototype construction - Information on available ECE funding to help offset certain prototype costs
July 7 (Tuesday)	4:00- 5:20pm	M3-1006	- Expectations for the Initial Prototype Demonstration - Expectations for the Progress Report and the revised Project Title & Abstract

COURSE DELIVERABLES AND GRADING

Deliverable	Dates and Locations	Grader	More information	Weight of Final Course Grade
Project Abstract & Project Agreement Form*	Uploaded to your LEARN Dropbox by 11:59pm on May 31	Course instructors	Page 10	10% (8+2)
Safety Quiz†	Must be done by 11:59pm on May 31 (LEARN)	Graded automatically by LEARN	Page 9	5%
Introductory video	Must be done by 11:59pm on May 31 (LEARN)	Course instructors	Page 9	5%
Project Specs. & Risk Assessment Document*	Uploaded to your LEARN Dropbox by 11:59pm on June 7	Course instructors	Page 15	15%
Detailed Design and Project Timeline Document*	Uploaded to your LEARN Dropbox by 11:59pm on July 05	Course instructors	Page 18	35%
Initial Prototype Demonstration*	Suggested time is July 21–26. Must be done before 5:00pm on July 28	Project consultant (fills in <i>Initial Prototype Demonstration Feedback Sheet</i>)	Page 22	Part of Progress Report grade
Progress Report and <i>Initial Prototype Demonstration Feedback Sheet</i> *	Uploaded to your LEARN Dropbox by 11:59pm on Aug 02	Course instructors	Page 22 Page 24	30%
Revised Project Title and Abstract	Uploaded to your LEARN Dropbox by 11:59pm on Aug 02	Course instructors		-5% - 0%
Reimbursement paperwork	Uploaded to your LEARN Dropbox by 11:59pm Aug 02	N.A.	Page 34	0%

Keep all your graded deliverables until March next year since you will need them to help write subsequent deliverables.

†Done by each student individually.

*One submission *per group*. Any group member may submit it. Except in unusual circumstances as determined by the course instructor (e.g., if one group member is obviously negligent in doing a fair share of work), all members of a design group receive identical grades on these deliverables.

All required documents must be submitted by the specified deadline. Submissions received after the deadline will incur a penalty, determined based on the number of days late, ranging from 5% to 30% of the assignment grade.

Rules about Group Work

You are expected to strongly collaborate with your fellow group members. Collaboration *between groups* in the form of sharing ideas, discussing design alternatives, debugging small fragments of source code, and debugging hardware designs is also encouraged. However, each group must do their own significant engineering design and analysis. All design/analysis content that has been shared or copied from another group must be clearly documented and acknowledged (including names of collaborators and the date of collaboration).

Appropriate Referencing of Materials

You are expected to cite any reference materials used in the development of your course deliverables. When referring to materials and using materials, references must comply with the formatting specified in the IEEE Citation Reference Guide (<http://www.ieee.org/documents/ieeecitationref.pdf>). In general, references should be provided for datasheets, white papers, research papers, books, lecture notes, and websites relevant to the design project if these materials have been used, in any way (e.g., using facts, explanations, equations, definitions, approaches, images,

code), in the production of the deliverable. However, do not reference your own work from previous ECE498A submissions. Note that the Final Report, which is part of ECE498B, will be electronically screened for plagiarism.

Use of Turnitin for Course Deliverables

Turnitin.com: Text matching software (Turnitin®) may be used to screen assignments in this course. Turnitin® is used to verify that all materials and sources in assignments are documented.

Details of the Safety Quiz

(See [page 8](#) for the deadline to complete the on-line Safety Quiz.)

Overview and Grading

Each student in ECE498A must individually complete the Safety Quiz that is on LEARN (go to the “SUBMIT” tab). The quiz is worth 5% of your final grade, and you need to pass the quiz to get access to the ECE project room (E2 -3353). You can attempt the quiz any time until the deadline stated on [page 8](#). You get up to 3 attempts, and the top score of the 3 attempts is recorded. You are given 90 minutes for each attempt.

If you somehow forget to do the quiz by the deadline, I will reopen the quiz on LEARN for you to try it; in such situations, however, your score will be reduced by 7.5 out of 15 points (i.e., a late penalty equal to 2.5% of your final grade) and a new deadline will be set. If you do not complete the quiz by the new deadline, your score will be zero.

Content of the Safety Quiz

In each attempt at the quiz, you will be asked 15 random questions, each of which is worth 1 point. Most questions are of true/false type but some are of multiple choice type. Each question deals with at least one of the following topic areas:

- E2-3353 safety rules (see the *Safety Rules for E2-3353* document)
- Electrical and mechanical safety (see the *Safety Rules for E2-3353* document)
- Basic electrical calculations (easily found on the web)
- Safety in general, including relevant legislation and the University’s Policies 34 and 60 (see [page 5](#))
- University Safety Office (see [page 5](#))
- Symposium safety (see [page 6](#))

Aids Allowed

During the quiz, you are not allowed to consult with another person (directly or indirectly), but you are allowed to use any documents, including those you find on the internet, to help you with the quiz. In fact, you are expected to look up information on the internet to correctly answer many of the questions.

Details of the Introductory video

(See [page 8](#) for the deadline to complete the on-line Safety Quiz.)

Introductory video: Overview and Content

Each group in ECE498A needs to prepare an introductory video (2-3 minutes). The video is for the general public, and you need to answer the following questions:

What is the problem you are trying to solve?

Who is your customer?

What is your proposed solution to the problem?

Example video: [SmartBrace Intro Video](#)

Details of the Project Abstract and Project Agreement Form

(See [page 8](#) for details about the deadline and method of submission for the Project Abstract and Project Agreement Form. See [page 28](#) for tips on forming a group, selecting a project topic, and finding a consultant.)

Project Abstract: Overview and Content

The Project Abstract is a concise document consisting of a project title and a single paragraph of text in which you describe the scope of your design project and its relevance. The following five components must be present:

Component	Comments
Project Title	Choose a title that is relevant to your project and is meaningful. The title must have no symbols (just letters and possibly numbers) and no acronyms ; the maximum length of the title, including spaces, is 50 characters.
Motivation	Briefly describe the context of the project and key background information. Make the motivation for your project specific, clear, and factual, as in the following examples: ▪ “Each year, about 850,000 vehicles in North America are involved in a collision due to failure to check one's blind spot.” (from Group 2009.54) ▪ “In the US and Europe 300,000 people die every year from cardiac arrest before they can reach medical care.” (from Group 2012.35)
Project objective	Clearly and explicitly state the objective of your project. Include a sentence that starts with something similar to “The objective/aim/goal of this project is to design...”. The objective must be open-ended enough to ensure that multiple solutions can be considered during the design process. (In other words, your project must be an <u>engineering design</u> project.) The project also must involve some aspect of electrical and/or computer engineering; it can be either a hardware-based, or a combination of hardware and software based projects.
Plausible design approach	Briefly describe a plausible design solution. As you work on your design throughout the term, you will very likely change or refine your design approach, but we want to make sure that you have at least one plausible approach in mind at the start of the project. Be sure that your design approach is clear (i.e., do not gloss over key design challenges with vague statements) and that it requires some <u>upper-year engineering knowledge</u> . Indicate explicitly what upper-year knowledge will be used (but do <u>not</u> mention course numbers such as ECE380).
Expected benefit of the design over existing alternatives	Clearly state at least one expected advantage of your design over existing or related designs. Be specific. For example: ▪ “The main advantage of this design over major alternatives is that physical size, strength, and/or mobility are not required to control the wheelchair.” (from Group 2011.43) ▪ “The Autotabber package can be used with existing guitar systems, and can detect different playing techniques on a per-string basis, which differentiates this product from pre-existing solutions.” (from Group 2012.31)

Note that you will resubmit your Project Abstract later in the term, so you will have an opportunity to change wording to reflect modifications to your project objective, design approach, etc. However, the grade assigned to your Project Abstract is based entirely on the first submission, so put in the effort now to create a well-crafted Project Abstract.

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Project Abstract: Format

The body of the Project Abstract should be:

- composed of a single paragraph on a single piece of paper
- no more than 200 words in length
- composed of pure text (no graphics, tables, or formulas; also include no references in an abstract)
- written in present (not future) tense since the abstract will be published after you complete your design
- well-written in plain professional English so that sentences are clear with a good flow from start to end
- free of jargon and acronyms as much as possible
- free of spelling and grammar mistakes.

Please be sure to list the names and student numbers of all group members with your submission.

Here is an example of a well-written abstract (based on that from Group 2016.11):

Group members: **First-name Last-name (20848325), email ID**
 First-name Last-name (20848326), email ID
 First-name Last-name (20848327), email ID
 First-name Last-name (20848328),email ID

Project title: SmartBrace

200-word Abstract: Every year, many athletes across the world suffer from sport injuries that can potentially lead to permanent physical damage and psychological impairment. The damage is more extensive for athletes who play at a professional level as it could result in a significant financial loss. Sport braces are commonly used to reduce the risk of injuries or to support the recovery process by providing pressure around a specific area. However, the added pressure often hinders an athlete's movement and does not feel comfortable. The objective of this project is to design a smart ankle brace that tightens up and applies pressure only when the athlete is at risk of injury. The SmartBrace uses a gyroscope and flex sensors to detect the onset of an ankle twist. The input to the system is collected from the athlete's movement, while the output of the system consists of a sudden tightening of the brace in the appropriate area. The required tightening force is calculated and applied using circuit design and control theory. The main advantage of this design over current alternatives is that it provides similar protection while being more comfortable and less restrictive to an athlete's movement.

Project Abstract: Other Examples

Go to LEARN to see hundreds of examples of Project Abstracts from previous terms. (**Caution:** The requirements for Project Abstracts change from year-to-year! Be sure you carefully follow the requirements on [page 10.](#))

Project Abstract: Grading

The Project Abstract is graded out of **20 marks** (and worth 8% of the course grade). Note that marks will be deducted for late submissions. Note that the 24 hour counting includes Saturdays, Sundays, and holidays.

Project Agreement Form

Groups need to fill in the yellow areas of the *Project Agreement Form*, shown on [page 14](#). A blank copy of the form is on LEARN. The Project Agreement Form is considered submitted only after all student members (4, 5, or 6 members) and the project consultant sign and date the form.

Submission of the *Project Agreement Form* is worth **5 marks** (and worth 2% of the course grade). Marks will be deducted for late submissions, as indicated near the bottom of the marking sheet shown above; beware that the 24 hour counting includes Saturdays, Sundays, and holidays. The *Project Agreement Form* must be submitted to receive a passing grade in ECE498A.

See [page 28](#) for tips on forming a group, selecting a project topic, and finding a consultant.

ECE498A/B: Project Agreement Form for Electrical and Computer Engineering Capstone Projects*

(For coordinator use only)

Group number: 2024.**Agreement Principles**

Intellectual Property Rights: All members of the University of Waterloo, including students, staff, and faculty members are subject to the terms of [Policy #73 on Intellectual Property Rights](#). This policy clearly states that ownership of the intellectual property rights created during the course of teaching or research activities belong to the creator(s). The creator(s) grant the University a non-exclusive, free, irrevocable license to copy and/or use such works in other teaching and research activities.

Student Academic Discipline: Students are expected to conduct their design projects in accordance with all university policies, procedures, and guidelines. Academic offences will be handled in accordance with [Policy #71 on Student Academic Discipline](#). This policy clearly states that students must not infringe unreasonably upon the rights of others, violate safety regulations, take part in plagiarism, or submit work previously submitted at this or any other academic institution. To ensure a safe working environment, all projects must be available for inspection by department staff.

Material Property: Components or material partially or wholly purchased using University funds belong to the University. Such material must be returned to the University at the end of ECE498B, at the discretion of the Course Instructor. Except where otherwise stated in this agreement, other components, material, or completed prototypes belong to the party that provided the source of funds used in purchasing them.

Consultant Responsibilities: The consultant acknowledges that his/her responsibilities include:

- being available to meet with the student team at mutually convenient times (nominally weekly) in this term and in the upcoming winter term to provide technical guidance to the student team,
- reviewing and commenting on (but not formally grading) the project deliverables as requested by students,
- assessing the prototypes demonstrated by students near the end of ECE498A and near the end of ECE498B, and
- upholding the students' academic interests above all others.

The consultant is a Professional Engineer, a University of Waterloo professor, a University of Waterloo lab instructor, or an individual approved by the ECE498A Course Coordinator. The consultant acknowledges the general terms of this agreement.

Student Responsibilities: Each student team member acknowledges that his/her responsibilities include the following:

- Each team member agrees to contribute positively, fairly, and professionally towards the completion of the design project.
- Each team member has full authority to represent the whole team in any matter relating to the project including the use of the project for student competitions or media interviews.
- Each team member is responsible for returning University property owed by any team member (see Material Property above).
- Each team member has reviewed [Policy #71](#) on Student Academic Discipline and [Policy #73](#) on Intellectual Property Rights.

Each student also acknowledges the general terms of this agreement.

Project title: **Autonomous Quadrotor Control System**

Student name (print neatly)	Student number (print neatly)	Student signature	Date signed
James Watt	20872832	<i>James Watt</i>	May 18 2023
Georg Ohm	20872847	<i>Georg Ohm</i>	May 18 2023
Michael Faraday	20872882	<i>Michael Faraday</i>	May 18 2023
Joseph Henry	20872890	<i>Joseph Henry</i>	May 18 2023
Consultant name (print neatly below)	Consultant email	Consultant signature	Date signed
Albert Einstein	genius@uwaterloo.ca	<i>Albert Einstein</i>	May 18 2023

*A blank Project Agreement Form is available on LEARN.

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Details of the Project Specifications and Risk Assessment Document

(See [page 8](#) for details about the deadline and method of submission for this document.)

Overview

Building on your Project Abstract, the Project Specifications and Risk Assessment document is a stand-alone technical report that describes your project specifications and assesses the risks that may jeopardize the success of your project.

Writing Style

This deliverable is a formal technical document. Hence, use formal language throughout. Use a professional-looking font and layout throughout. Standard technical writing habits should be followed:

- units of measurement (e.g., “m” for meters, “V” for volts) should be indicated for every numeric quantity
- acronyms and jargon should be defined the first time they are used
- each figure should have a number and caption below it, and each figure should be cited in text
- each table should have a number and caption above it, and each table should be cited in text
- figure/table numbering should be consistent in the report body
- for figures/tables taken from another source, cite that source by reference number in the caption
- all figures/tables should appear in the report close to where they are first cited in the main body
- all figures/tables must be readable and should not be split over multiple pages if possible
- each reference (in the References section) should be cited at least once in the report body.

In addition, pay attention to structure and flow, as well as sentence and paragraph structure. Use acronyms and jargon sparingly. Spelling and grammar should be correct. Writing quality is very important: you might have the greatest design project in the class, but if you do not explain effectively and efficiently what you have done, the reader will not appreciate, and possibly not even understand, what your project is about.

Throughout your writing, keep in mind that the course instructor does not have expertise in every subfield of electrical and computer engineering. Hence, if the scope of your project is very different from his expertise, you would be wise to take extra care in minimizing jargon and explaining specialized background. Feel free to ask the instructor how knowledgeable he is with your project area.

Before writing anything, you should carefully study the marking sheets/rubrics.

Page Limit, Font Size, Font Spacing, and Margins

To encourage you to be concise, and to keep the marking load on the course instructor realistic, the page limit is **10 pages**, including everything (title page, table of contents, ..., references).

To maximize readability of your report, please use an 11 or 12 point font and 1.15 to 1.50 line spacing, with at least 1-inch (2.54 cm) margins on all edges; use at least a 9 point font for words that appear in figures. Any standard professional-looking font is acceptable.

Required Components of the Project Specifications and Risk Assessment Document (10 pages max)

Component	Comments
Title page	Include on the title page: - the university, Faculty, and department names (University of Waterloo, Faculty of Engineering, Department of Electrical and Computer Engineering) - your project title and your <u>group number</u>* (e.g., 2024.75) - a list of the group members, including student numbers - the name of your consultant - the submission date
Table of Contents	Include a Table of Contents on a separate page. Number all pages after the title page.
1. High-Level Description of Project	
1.1 Motivation	- Expand on the motivation included in your Project Abstract. <u>Include at least one reference</u> to justify any motivating facts or claims.
1.2 Project Objective	- The project objective must be clear and explicit. It must also be open-ended with more than just one obvious solution. Take from your Project Abstract and revise if necessary.
1.3 Block Diagram	- Provide a well-labelled block diagram of the project's inputs, subsystems, and outputs. Since you have not yet done your detailed design, the subsystems of the block diagram that are associated with parts that you have to design will be non-specific (e.g., it's appropriate to label a subsystem as "position sensor" rather than a specific type of position sensor if the selection of that sensor is part of your design project). If you know that a particular part will be used in your project (i.e., you are not going to be designing that part), then you should label the part specifically in your block diagram. We recognize that your block diagram may change over the term as you work on your design project. A typical block diagram will have 4 to 6 subsystems included. - Use text and possibly additional figures to clearly explain how your system works. - <u>Clearly indicate on your block diagram which subsystems you will be designing, and which ones you will not be designing.</u>
2. Project Specifications	- Write in <u>table form</u>, with additional paragraphs to help explain your specs as needed.
2.1 Functional Specifications	- Clearly label each specification as either <i>essential</i> or <i>non-essential</i>.
2.2 Non-functional Specifications	- The set of specs should be verifiable, open-ended, appropriate, complete, & challenging. - See the detailed comments on the next page.
3. Risk Assessment	In <u>table form</u>, identify the 3 (or more) most significant situations that could prevent successful completion of the project. For each situation, specify: - the nature of the situation (time management problems, group dynamics problems, insufficient skill or knowledge of group members, insufficient funds or time to purchase parts, insufficient access to needed tools/equipment, technical risks such as insufficient battery power or processing power, risk of practice not matching theory, ...) - the potential impact of the situation (e.g., high, medium, or low impact) - the probability of the situation occurring (e.g., high, medium, or low probability). For situations that have high impact and/or high probability, specify both: - what can be done to reduce the likelihood that the situation occurs - what can be done to reduce damage/harm caused if the situation does occur.

References

Follow IEEE formatting and number references with [1], [2], etc. You need to reference any datasheets, figures, all non-obvious facts/material taken from anywhere else, books, websites, etc. Do not reference your own work from previous ECE498A submissions.

*Your group number is written on the top of your *Marking Sheet for Project Abstract and Project Agreement Form*. A complete list of project numbers is also available on LEARN.

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Regarding the Project Specifications part of the report (Section 2):

(a) Specifications are traditionally classified as functional or non-functional (although sometimes the distinction is fuzzy, so use your best judgement):

- A functional specification* is a design specification that directly relates to the operation/behavior of the system, i.e., it deals with what the system is supposed to do. For example, a project involving the automation of a quadcopter system might have a functional specification stating “The quadcopter must be able to fly autonomously for at least 30 minutes.” or perhaps “The quadcopter must be able to hover 2 meters off the ground with no more than 5 cm deviation in any direction and no more than 10 degrees rotation in any direction.”

You need to decide if each functional specification is either essential or non-essential. Essential specifications are those that your project must meet for your design to be considered satisfactory. Non-essential specifications are less critical to the basic operation of your system, but by meeting them, your project will be more than just satisfactory. Do not include specifications that you have no hope or intention of meeting.

- A non-functional specification* is a design specification that does not directly relate to how the system operates. Rather, non-functional specifications describe characteristics of the system (typically things like size, mass, appearance, cost, reliability, maintainability, usability). For the quadcopter project, an example of a non-functional specification might be “The designed control system for the quadcopter must fit within a 3 cm by 5 cm by 5 cm box.”

Non-functional specifications should also be listed as essential or non-essential.

(b) A good set of specifications is:

- Verifiable: Every specification must be verifiable, i.e., it must be possible to *objectively* determine if it is satisfied or not. Avoid vague specifications, e.g., “The quadcopter control system must be light-weight.” Quantify where possible, e.g., “The mass of the quadcopter control system must be under 300 g.”
- Open-ended: Each specification should relate to parts of the project that you are designing, and therefore it needs to be open-ended. For example, “The quadcopter altimeter should be the SuperBrand X Altimeter-Z99.” is not suitable as a specification for your ECE498A/B design project because the spec is not open-ended. But something like “The quadcopter control system must sense altitudes between 0 and 5 m, to within 5 cm accuracy.” is suitable (but only if you are planning to include the choice of altitude-measuring device as part of your design; if you know you will use the SuperBrand X Altimeter-Z99, then indicate this in your block diagram and do not create a specification related to the altimeter).
- Appropriate: Do not include specs that are irrelevant or that are actually detailed design decisions.
- Complete: Be sure that there is at least one specification associated with each subsystem that you will be designing. Moreover, be sure that all the important aspects of the system behaviour are captured by the specifications.
- Challenging: At least some of the specifications must be challenging (but still realistic) for a group of fourth-year engineering students. For example, “The quadcopter must be able to hover 2 meters off the ground with no more than X cm deviation in any direction.” probably has a good level of challenge for $X = 5$, but it is not challenging enough if you use $X = 100$, and it is likely too challenging if you use $X = 0.1$.

*Engineers sometimes like to distinguish between specifications and high-level (often vague) customer-focused needs, called *requirements*. In practice, it can be a huge effort to figure out customer requirements. Once the requirements are known, more concrete (and verifiable) *specifications* are derived. For your project, we want you to jump right to the specifications since you don't have a real customer driving your project.

Grading of the Project Specifications and Risk Assessment Document

The Project Specifications and Risk Assessment document is graded out of **50 marks**. Note that, for late submissions, the 24 hour counting includes Saturdays, Sundays, and holidays.

Details of the Detailed Design and Project Timeline Document

(See [page 8](#) for details about the deadline and method of submission for this document.)

Overview

Building on your Project Specifications and Risk Assessment document, the Detailed Design and Project Timeline document is a stand-alone technical report that describes your detailed design and provides a timeline for the completion of your project.

Writing Style, Page Limit, Font Size, Font Spacing, Margins, Printing and Binding

Follow the instructions for the Project Specifications and Risk Assessment submission (see [page 15](#)), except the page limit for the Detailed Design and Project Timeline document is **30 pages**, including everything (title page, table of contents, ..., references, optional appendices).

Exception to line spacing requirement: Single-line spacing may be used for the Table of Contents and References.

Note that almost all the documents should be written in present tense (e.g., use “Power for the circuit is provided via connector A.” instead of “Power for the circuit will be provided via connector A.”).

Required Components of the Detailed Design and Project Timeline Document

See the table on the next page.

Level of Detail in Section 3 of the Report

The most important part of this document is Section 3, where you describe your detailed design. For each subsystem (or possibly group of subsystems) that you designed, you need to provide two types of information:

1. Describe the details of the final design of that subsystem and argue that it satisfies the design specifications. You want to include enough detail of your final design that ideally another group could, in theory, use your documentation to construct your prototype. Hence, it is appropriate to provide circuit schematics, PCB layouts, pseudo-code (or even code fragments), UML diagrams, flowcharts, state diagrams, ER diagrams, data flow diagrams, timing diagrams, CAD drawings, part listings, etc.
2. Clearly describe in Section 3 the engineering design process you followed that led to your final design (i.e., explain the design iterations and/or alternatives that you considered). You will be assessed on how effectively you justify your design decisions in the engineering design process, and on your use of quantitative technical analysis. You want at least some of your design and/or technical analysis to be at an advanced level, requiring substantial upper-year engineering knowledge.

Quantitative Technical Analysis (QTA)

Quantitative Technical Analysis (QTA) – Demonstrates:

- **Data-Driven Approach:** QTA relies heavily on data analysis. A solution that has used QTA will likely have been developed based on thorough data collection, processing, and analysis. Must include references to specific datasets, measurements, or simulations that were used to inform design decisions (prior to starting solution).
- **Quantifiable Metrics:** QTA involves the use of quantifiable metrics to evaluate the performance, efficiency, or effectiveness of a solution. Therefore, the solution should include numerical values, performance benchmarks, or comparative analyses to demonstrate the effectiveness of the proposed approach based on your unique situation.
- **Optimization Techniques:** QTA involves optimization techniques aimed at maximizing or minimizing certain objectives within specified constraints. Solutions developed using QTA may include references to optimization algorithms, parameter tuning processes, or sensitivity analyses to achieve the desired outcomes.
- **Mathematical Models:** QTA frequently employs mathematical models to describe and analyze complex systems or phenomena. A solution that incorporates QTA will include mathematical equations, algorithms, or simulations to model the behavior of the system under study and facilitate analysis and optimization.
- **Rigorous Analysis:** Solutions developed using QTA are typically characterized by a rigorous analytical approach. This may involve mathematical derivations, statistical analyses, or computational simulations to support design decisions and validate the proposed solution.
- **Evidence-Based Conclusions:** QTA emphasizes evidence-based decision-making. Therefore, a solution that has utilized QTA will likely present conclusions and recommendations supported by empirical evidence, quantitative analysis, and logical reasoning rather than anecdotal evidence, intuition or a decision matrix.
- **Iterative Process:** QTA often involves an iterative process of refinement and improvement based on feedback and analysis. Therefore, the solution may demonstrate evidence of multiple iterations, sensitivity analyses, or trade-off studies aimed at iteratively improving the design or performance of the solution. Document all processes and include solutions and results from previous iterations.

Quantitative Technical Analysis (QTA) – Constraints:

- Unique to your project
- Can't present cited work/data as your QTA. For example, copying well-known formulas and stating you used it in your design does not count as QTA. You need to show how and why you used the formulas and what your calculations are. They calculations should also involve substantial 3rd and 4th year concepts.
- Trivial calculations do not count as QTA. For example, including Ohm's Law calculations to determine the resistance value of a current-limiting resistor is not QTA
- Can't evaluate the tools you are using for your project. For example, citing speed of access for data in a database for various different database topologies does not count as QTA for your project as you didn't do the measurements.

If you find that you are running out of room in your report, focus on the most interesting or challenging design material and emphasize those subsections that have the most substantial engineering design and/or quantitative technical analysis.

Required Components of the Detailed Design and Project Timeline Document *

Component	Comments
Title page	Include on the title page: - the university, Faculty, and department names (University of Waterloo, Faculty of Engineering, Department of Electrical and Computer Engineering) - your project title and your group number - a list of the group members, including student numbers - the name of your consultant - the submission date
Table of Contents	Include a Table of Contents on a separate page. Number all pages after the title page.
1. High-Level Description of Project 1.1 Motivation 1.2 Project Objective 1.3 Block Diagram	- Extract these items from your Project Specifications and Risk Assessment submission, updated to reflect the instructor's feedback and any other changes you have made. - Be sure that you clearly indicate on your block diagram which subsystems you designed as part of your project.
2. Project Specifications 2.1 Functional Specifications 2.2 Non-functional Specifications	- Extract from your Project Specifications and Risk Assessment submission, updated to reflect the instructor's feedback and any other changes you have made. [Talk to the course instructor before making any major changes to the specs.] - Specifications must be well organized in table format.
3. Detailed Design 3.1 Subsystem X Design 3.2 Subsystem Y Design 3.3 ...	Describe your detailed design: - Use a separate subsection (3.1, 3.2, etc.) for each subsystem (or possibly group of subsystems) you designed; these subsystem names should match up with those on the Block Diagram. - Give a detailed description of your final design for the subsystem. - Explain clearly why/how your design satisfies the project specifications. - Explain clearly the design iterations and/or alternative designs that you considered - Clearly justify (i.e., explain the rationale for) all significant design decisions related to your design iterations and/or alternatives - Include quantitative technical analysis (engineering theory and/or simulations, as appropriate) to help justify your design decisions and/or to evaluate your design - You want at least some of your design/analysis to use upper-year knowledge
4. Discussion and Project Timeline 4.1 Evaluation of Final Design 4.2 Use of Advanced Knowledge 4.3 Creativity, Novelty, Elegance 4.4 Student Hours 4.5 Potential Safety Hazards 4.6 Project Timeline	You must do at least the following: - Summarize how effectively your final design meets the project objective & specs - Summarize where upper-year ECE technical knowledge was used in analysis or design - Highlight the most creative, novel, and/or elegant aspects of your design - Indicate the total hours of work <u>each</u> student has put in so far (Hint: consult your logs) - Identify any potential safety hazards that you should think about as you build your prototype (recall the information on pages 5 and 6) - Present a realistic timeline (displayed as a <u>table</u> or a <u>Gantt chart</u>) showing how you plan to finish your project. Include at least the following activities in your timeline: - o revision/refinement of your design if necessary (any revisions will be submitted as part of the Final Report in ECE498B) - o prototype construction (order parts, shipping delays, construction, testing) - o demonstration of your prototype progress at the end of ECE498A (Note: your prototype is to be <u>at least 50% (ideally 75%) built</u> by end of ECE498A) - o preparation of your Final Report (deadline is in mid-March as part of ECE498B) - o demonstration of your final prototype (deadline is mid-March in ECE498B) - o preparation for the design symposium (held in mid-March in ECE498B)
References	Follow IEEE formatting and number references with [1], [2], etc. Do not reference your own work from previous ECE498A submissions.
Appendices (optional)	Supporting details that are not essential to the report can be put in appendices.

*Maximum of 25 pages.

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Grading of the Detailed Design and Project Timeline Document

The Detailed Design and Project Timeline document is graded out of **80 marks**. Note that marks will be deducted for late submissions, as indicated on the marking sheet; beware that the 24 hour counting includes Saturdays, Sundays, and holidays.

Details of the Initial Prototype Demonstration and Progress Report

(See [page 8](#) for details about the suggested deadline for the Initial Prototype Demonstration, and for details about the deadline and submission method for the Progress Report.)

Overview

The Initial Prototype Demonstration is where you meet with your consultant and show him or her how your prototype construction is progressing: you want your prototype to be at least 50% (ideally 75%) done by the demonstration date. The consultant provides some written feedback.

The Progress Report is a brief document that allows you to formally and honestly report to the course instructor how your project is progressing. The report includes the feedback that your consultant provided at the Initial Prototype Demonstration, your student logs, and discussion on the progress that you have made. You also have an opportunity to revise your Project Timeline.

Logistics of the Initial Prototype Demonstration

1. Arrange with your consultant a mutually-convenient time and location for your demonstration. A typical demonstration takes about 30 minutes. The demonstration must take place before the Progress Report submission deadline. You would be wise to communicate with your consultant at least several weeks before the deadline. It is best if all group members attend the demonstration, but it is acceptable if one person is away (due to illness, time conflicts, etc.).
2. Confirm, 24 hours ahead of time, the time and location of your appointment with your consultant.
3. Ensure that you bring to the demonstration the following items:
 - a. Your prototype and any other equipment required to demonstrate how your prototype is progressing (e.g., a laptop, power supply, batteries, ...).
 - b. A hardcopy of your Block Diagram (extracted from Section 1.3 of the Detailed Design and Project Timeline document) so that you can explain your design to your consultant.
 - c. A hardcopy of your Project Specifications (extracted from Section 2 of the Detailed Design and Project Timeline document) so that you can explain what your prototype should do once it is complete.
 - d. A blank hardcopy of the Initial Prototype Demonstration Feedback Sheet* shown on [page 23](#).
4. When the demonstration begins, describe the project objective and give an overview of your design. Show your consultant the progress you have made on your prototype construction.
5. The consultant, working with the *Initial Prototype Demonstration Feedback Sheet*, will provide you with feedback. Be sure your consultant signs the *Initial Prototype Demonstration Feedback Sheet*.
6. You should include the signed *Initial Prototype Demonstration Feedback Sheet* in Appendix B of the Progress Report (see pages [24-25](#)), to be submitted by the deadline stated on [page 8](#).

*A blank *Initial Prototype Demonstration Feedback Sheet* is available on LEARN.

Consultant Feedback at the Initial Prototype Demonstration

Your initial prototype is not formally graded by your consultant. Rather, your consultant provides you with some honest feedback via the *Initial Prototype Demonstration Feedback Sheet*, which is shown below. This feedback affects the grade of your Progress Report (see page [22](#)). Note that in ECE498B your final prototype will be formally graded by your consultant and then again as part of your presentation at the March design symposium.

ECE498A: Initial Prototype Demonstration Feedback Sheet				
Group number: 2024. _____				
Instructions for consultant: By checking the most appropriate boxes below, please provide <u>frank feedback</u> to the student group about how you think their project is progressing. At this point in the term, students should have finalized their designs and their prototypes should be at least 50% (preferably 75%) complete. Students will respond to this feedback in their Progress Report. In March next year, the students will approach you again to schedule a Final Prototype Demonstration, in which you formally assess (and assign grades) on their project and prototype.				
Assessment of the level of challenge and upper-year knowledge used in the project				
☐ (4) Project is significantly challenging, clearly requiring substantial 3 rd or 4 th year engineering knowledge				
☐ (3) Project is challenging, arguably requiring substantial 3 rd or 4 th year engineering knowledge				
☐ (2) Project is fairly straightforward, requiring substantial 2 nd year university-level engineering knowledge				
☐ (1) Project is simple, requiring little or no university-level engineering knowledge beyond 1 st year				
Assessment of how complete the prototype construction is				
☐ More than 90% done	☐ 50% to 74% done	☐ Less than 25%		
☐ 75% to 90% done	☐ 25% to 49% done	☐ Impossible to judge		
Your confidence that, by next March, the prototype will be 100% complete and satisfy all <u>essential</u> design specs				
☐ Very high	☐ High	☐ Unsure	☐ Low	☐ Very low
Your confidence that the group is working well as a team				
☐ Very high	☐ High	☐ Unsure	☐ Low	☐ Very low
Your confidence that the group has put appropriate effort and time (ideally 120 hours <u>per student</u>) into the project				
☐ Very high	☐ High	☐ Unsure	☐ Low	☐ Very low
Other feedback to the students:				
Consultant's signature: _____ Date: _____				

Writing Style, Page Limit, Font Size, Font Spacing, Margins, Printing and Binding of the Progress Report

Follow the instructions for the Project Specifications and Risk Assessment submission (see [page 15](#)), except the page limit for the Progress Report is 6 pages, including everything *other than the appendices*.

Required Components of the Progress Report (6 pages max, *excluding appendices*)

Component (start each on a new page)	Comments
Title page	Include on the title page: - the university, Faculty, and department names (University of Waterloo, Faculty of Engineering, Department of Electrical and Computer Engineering) - your project title and your group number - a list of the group members, including student numbers - the name of your consultant - the submission date
1. Overview of Project 1.1 Revised Project Abstract 1.2 Original Project Timeline	To help remind the instructor about your project, include here: - your final Project Abstract (this is the same revised Project Abstract that you must submit on LEARN at the end of term—see page 25) - your Project Timeline, extracted from Section 4.2 of the Detailed Design and Project Timeline Document.
2. Current Status of Project 2.1 Prototype Completion 2.2 Student Hours	In this section: - Provide an estimate of how complete your prototype construction is, expressed as a percent. You must convincingly justify your number; do not just state a number. - If your estimate differs from that of the consultant (see Appendix B), provide an explanation. - State how many hours <u>each</u> student has put into the project, and then sum the hours to get the total number of hours the group has put in. These numbers must agree with the student logs in Appendix A. - If your hours are less than the expected 120 hours per student, or if your hours are inconsistent with the consultant's impression (see Appendix B), or if there is significant variation in the number of hours put in by the group members, provide an explanation.
3. Discussion	In this section: - If you and/or your consultant estimate that your prototype is less than 50% complete, explain what changes you will make to get your project on schedule. Changes might include changes to time management, changes to how the group functions, or changes to project specifications. If you revise your Project Timeline, include the new one here. - Express how confident the group is that your prototype construction will be 100% done, and meet all essential design specifications, by the March symposium in ECE498B. Justify your level of confidence. - If your consultant indicates that your project is not very challenging (see Appendix B), indicate if you intend to make it more challenging. A large portion of the points for your Final Prototype Demonstration (in ECE498B) is related to the consultant's impression of the level of challenge of your project. In addition, some of the Final Report marks (in ECE498B) relate to the level of difficulty of your project.
Appendix A: Student Logs	- Include the logs for all students in your group (see page 27). - Student must sign their logs and include the "By signing above..." statement on page 27. - Please do not submit full lab books as logs since they are too bulky and heavy; if you used a lab book and want to submit it as evidence to support your log entries, then please photocopy your pages and submit those instead.
Appendix B: <i>Initial Prototype Demonstration Feedback Sheet</i>	Put in this appendix the signed Initial Prototype Demonstration Feedback Sheet from your Initial Prototype Demonstration (see page 22).

Grading of the Progress Report

The Progress Report document is graded out of 30 marks. Note that marks will be deducted for late submissions, as indicated on the marking sheet; beware that the 24 hour counting includes Saturdays, Sundays, and holidays.

Details of the Revised Project Title and Abstract and Revised Design Documents Submission

(See [page 8](#) for details about the deadline and method of submission for the revised Project Title and Project Abstract.)

Overview

The project title and Project Abstract submitted in earlier documents probably do not accurately reflect your final design. You now have an opportunity, right at the end of term, to modify the title and refine the Project Abstract, taking into account any feedback provided from the course instructor and any changes in your project. The project title and Project Abstract that are submitted at the end of this term are those that will be used in ECE498B to advertise the March design symposium. They will also be posted publicly on the Faculty of Engineering's web page and will be sent to media sources. Hence, it is in your interest to make the title and abstract as accurate, polished, and interesting as you can. Minimize jargon and acronyms in your Project Abstract. Even if you do not want to make any changes to your project title and abstract, you need to re-submit them as described below so that I have a softcopy of these items.

We will publish an online brochure at the symposium that has a photograph of each group and a shortened (85-word maximum) Project Abstract. So you also need to submit a shortened Project Abstract and a list of the names of group members, spelled out exactly as you want them to appear in the brochure and on the symposium name tags. The shortened abstracts will be given to potential markers for them to choose projects for marking at the symposium.

Formatting and Content

This document is to be submitted as a softcopy on LEARN. Please submit a **single doc or docx file (NOT a pdf file** since cutting and pasting doesn't work well with pdf's) containing all the following items:

1. your group number
2. a list of names of group members and student numbers
3. your revised Project Title (text only, no symbols, at most 50 characters)
4. your revised Project Abstract (single paragraph, max 200 words, pure text, present tense, no graphics or tables or formulas, no references, minimal jargon and acronyms, no spelling or grammar mistakes)
5. a shortened Project Abstract for the brochure (single paragraph, max 85 words, pure text, present tense, no graphics or tables or formulas, no references, minimal jargon and acronyms, or no spelling mistakes).

Marking

Any group member may submit the document on LEARN on behalf of your group. Marking is out of **(-5 – 0) marks**, as follows:

- | | |
|--|------------|
| ▪ If you submit by the deadline and with all content and correct formatting: | 0 mark |
| ▪ If you submit by the deadline but with formatting problems: | -2.5 marks |
| ▪ If you missed the deadline: | -5 marks |

EXPECTED COURSE WORKLOAD AND STUDENT LOGS

As with all ECE courses, it is expected that each student puts in about 120 hours of work in each of ECE498A and ECE498B. It is important to keep the expected workload in mind when choosing a project, coming up with a design, and planning your time throughout the term. A common mistake is that groups spend too little time on their project in the first half of the term, putting themselves under great pressure over the remainder of the term.

To help you ensure that you are spending the appropriate amount of time on your project throughout the term, **each student needs to keep a personal log of their activity**. The logs are to be submitted as part of the Progress Report due at the end of the term (see deadline stated on [page 8](#).)

The following is an example of a partially-filled in log. Please use this format for your log. Logs can be typed or handwritten. You must include the “By signing above, I am stating that this is an accurate account of the tasks, dates, and times that I worked on my capstone design project.” statement in your log, and it must be signed when submitted at the end of term as part of the Progress Report.

ECE498A: Student Log*					
Name: <u>Batman</u> Group: <u>2026.75</u> Signature: <u><i>Batman</i></u>					
By signing above, I am stating that this is an accurate account of the tasks, dates, and times that I worked on my capstone design project.					
Task	Date	Start time	Finish time	Hours	Running total of hours
Formed group with A.J. and D.M. and R.Y. Brainstormed topic and converged on 3 ideas.	May 7	8:30am	10:00am	1.5 hours	1.5 hours
...					
Finalized project abstract and agreement form	May 19	Noon	1:00pm	1.0 hours	12.5 hours
...					
Worked on specs document with team	May 22	8:00pm	9:00pm	1.0 hours	19.5 hours
...					
Finished specs document	May 30	8:00pm	10:00pm	2.0 hours	30.0 hours
Researched amplifier design methods	May 31	9:00pm	10:30pm	1.5 hours	31.5 hours
...					
Proofread and submission of progress report	July 28	9:00pm	10:30pm	1.5 hours	120 hours

*A blank *Student Log* is available on LEARN.

FORMING A GROUP, SELECTING A PROJECT TOPIC, AND FINDING A CONSULTANT

Although other sequences are possible, we recommend that you first form a group, then select a topic, then lastly find a consultant.

Forming a Group

- All ECE design project groups need to have 4-6 members to be called a complete group. A group with more than six members or less than four in it will not be permitted. Note that project expectations are higher for larger team sizes. If your group ends up shrinking (e.g., a group member fails 4A), the project must continue with a reduced group size.
- Students in a group may be from Electrical Engineering or Computer Engineering or a mixture of the two. If you want to have a group with non-ECE members, talk to the Course Coordinator immediately.
- Group members work together extensively, so be sure you choose teammates who you get along with. Also be sure you can find time at least twice a week where the whole group can meet together.
- Be sure your team has a good balance of professional skills: you definitely want at least one person who can write well, someone who can present well, someone who is extremely organized, and someone who can encourage and lead the team. Time management is critically important for open-ended projects.
- The greater the breadth of technical skills (programming, analog circuit design, digital circuit design, computer interfacing, control, power systems, soldering, basic mechanical design, basic parts assembly, ...) in your team, the greater the flexibility you have in choosing a project topic.
- If you are having trouble finding a group or if your group needs more members, you can use the LEARN discussion group to communicate with other students. If you are not part of a complete group by the second lecture, come to the class for the second lecture (see [page 7](#)) to meet others who are in the same situation.

Selecting a Project Topic

- Recall from [page 2](#) that there are four guiding principles to keep in mind when choosing a project topic and the specific project objective:
 - the project must involve engineering design (i.e., a creative, iterative, and open-ended process),
 - it must involve some quantitative technical analysis (i.e., quantitative investigation using engineering theory, simulations, and/or experimental data),
 - **It must involve at least 2 of the core electrical/computer engineering components in a substantial way: communications, digital circuits, electronic devices, circuits, programming, power systems, RF, controls, and**
 - it must be at an advanced level (i.e., require upper-year engineering knowledge for design and/or analysis and be complicated enough to require, over the two terms, about 960 hours of work for 4-group teams, or 1200 - 1440 hours of work for 5/6-group teams).

Much of the project grading focuses on these issues.

- The project should match the group composition:
 - Be sure that your group has the technical skills necessary to work on the project you select; if your project involves extensive analog circuit design, for example, then at least one or two members should be very strong in this area.

- Everyone in the group should be enthusiastic about your project! Non-enthusiastic members won't be willing to put full effort into the project.
- Here is a process that can be used to choose a topic:
 1. Brainstorm to generate a list of possible topics—see the resources below to help with brainstorming. Any topic related to electrical or computer engineering is allowed.
 2. Narrow down the list of potential topics to a short list that everyone in the group likes. Remove or scale-down topics that are obviously too expensive or too challenging.
 3. Do some quick research on each of the short-listed topics: for each one, come up with one or two specific project objectives. Each potential project objective should be well motivated by a societal need. Your group should have some idea on how you can come up with a solution that is better (in at least one way) than existing solutions. Remember to keep in mind that your design project needs to involve genuine engineering design, and must be at an advanced level. Also keep in mind your team's skills.
 4. Narrow down the topic and project objective to just one, perhaps by having everyone in the group rank the short-listed topics.
 5. Iterate until your project objective is:
 - unambiguous—the main goal should be clear
 - challenging—recall that you need to use upper-year knowledge in your project, and it should not be possible to come up with a design and build a prototype in a weekend
 - open-ended—a good design project does not have just one obvious solution; by making your project objective open-ended, you ensure that you will need to make non-trivial decisions and that your design solution can be iteratively improved
 - realistic—you need to have your detailed design done by the end of June and your prototype mostly done by the end of term (see deadlines on [page 8](#)), so keep the design project realistic.
 6. You are now in a great state to write up your Project Abstract (see [page 10](#)).
- Here are some resources to help you brainstorm ideas for topics:
 - On LEARN you will find hundreds of abstracts for ECE projects from previous years. Obviously, you should not copy someone else's project, but by looking at their abstracts you may get new ideas.
 - On LEARN you will find a few specific project ideas suggested by faculty or others.
 - Think “outside the box” when brainstorming: spend time thinking of broad areas such as robotics, communication systems, automotive systems, entertainment/gaming, appliances, manufacturing/assembly systems, health care, internet-of-thing systems, smart phones, human-machine interaction, smart grid systems, smart home systems, energy systems, environmental systems, sports/exercise equipment, transportation systems, vending machines, security/military systems, just to name a few.
 - Read news headlines to see what societal needs exist, and think about how technology might be able to deal with those needs.
 - Ask friends and non-technical people about what societal needs they see.
- Avoid making the mistake of selecting a design solution before you choose the particular project objective! It is important to have some confidence that you can come up with a design and working prototype, but the focus here is defining the objective you aim to meet. The details of the design will follow afterwards.
- If you are having trouble converging on a topic, feel free to contact the Course Instructor for help.

Finding a Consultant

After your group has a topic (even if the specific project objective is not yet 100% finalized), you should search for a project consultant. The consultant's main role is to help groups, upon request, with technical questions and to assess the Initial Prototype Demonstration (in ECE498A) and the Final Prototype Demonstration (in ECE498B). **You need to have a consultant to get credit for ECE498A.**

The consultant should be:

- a University of Waterloo professor (any engineering department),
- a University of Waterloo lab instructor (any engineering department),
- a professional engineer (whether or not they are affiliated with UWaterloo), or
- some other individual pre-approved by the Course Coordinator.

ECE professors and lab instructors are informed early in the term that groups of students will be looking for consultants, so do not be afraid to approach them. The different research areas of the ECE department can be found in the following link:

[Research Fields | Electrical and Computer Engineering | University of Waterloo](#)

A directory of ECE Faculty is listed in the below link. Only contact faculty/lecturers, lab instructors, or other technical individuals. Please do not contact strictly admin personnel, unless you know they have the technical expertise that you require for your project.

[Contacts | Electrical and Computer Engineering \(uwaterloo.ca\)](#)

A directory of MEC/MTE Faculty is listed in the below link. Only contact faculty/lecturers, lab instructors, or other technical individuals. Please do not contact strictly admin personnel, unless you know they have the technical expertise that you require for your project.

[Contacts | Mechanical and Mechatronics Engineering \(uwaterloo.ca\)](#)

Note that groups may not have more than one official consultant; anyone else who helps you with your project can be acknowledged in your reports.

Once you have a consultant, the Project Agreement Form (see [page 14](#)) needs to be filled out. The deadline for submission is given on [page 8](#). Note that marks are deducted for late submissions (see marking sheet on [page 12](#)).

Key tips for working as a group

- Assign specific responsibilities to every group member. Without this individual accountability, projects usually get way behind schedule. Do not say that “everyone” is responsible for X or Y in a project.
- Communicate with your group members frequently. If you think something isn't going well, talk about it early before it becomes a serious problem. Remember to attack issues, not people.
- Come up with a method for making decisions, and everyone should be prepared to compromise at least sometimes.

TIPS FOR PROTOTYPE CONSTRUCTION

Overview

The construction of a working prototype is a core part of the ECE Capstone Design Project. An Initial Prototype Demonstration takes place near the end of this course, and a full Final Prototype Demonstration takes place in ECE498B. Prototypes are also evaluated during the design symposium held next March.

It is expected that your prototype be at least 50% complete for the Initial Prototype Demonstration near the end of this term. Although it makes sense to first work out all the design details and then build the prototype, in cases where hard-to-buy items need to be purchased for prototype construction, groups should order components/equipment early in the term because shipping delays can be substantial.

Buying Small Parts on Campus

You can buy many small parts from RidgidWare, run by the Engineering Society. See details at <https://www.engsoc.uwaterloo.ca/services/ridgidware/>.

Borrowing other ECE Equipment

The ECE Project Room (E2-3353) is equipped with standard power supplies and instrumentation that you can use. If you need other equipment (e.g., a high-capacity power supply), talk with one of the course instructors.

Where to Buy Components/Equipment off Campus

If you cannot find the desired equipment using the methods described above you will have to buy the parts off campus.

Buying as locally as possible is usually preferable since time-consuming and expensive shipping is avoided. If you buy on-line, it's usually faster and cheaper to use a Canadian-based supplier. As a last resort, order from the USA. Do not order any 120V AC parts that do not meet safety standards; those will not be allowed at the symposium next March. The table on the next page lists some suitable vendors. Be sure you sign into the Canadian site of the on-line vendors.

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Vendor	Location	Typical equipment/components (subject to change)
<i>The Source</i> 50 Westmount Road, Waterloo	Walkable or can order on-line	Simple electronics, computers, audio, GPS, cables, batteries
<i>Glenbriar Home Hardware</i> 262 Weber St. N., Waterloo	Walkable	Hardware (screws, hinges, ...), building supplies, plumbing, tools, paint, electrical components, adhesives
<i>Canadian Tire</i> 400 Weber St. N., Waterloo	Walkable	Hardware, electrical components, tools, sports equipment, paint, small appliances, auto and engine parts, adhesives
<i>Canada Computers & Electronics</i> 230-234 Weber St. N, Waterloo	Walkable or can order on-line	Computer hardware, phones, cameras, cables
<i>Home Depot Canada</i> 600 King St. N., Waterloo	Driveable or can order on-line	Hardware, building supplies, tools, wood/lumber, electrical components, paint, plumbing, large appliances, adhesives
<i>Lowe's Canada</i> 345 The Boardwalk, Waterloo	Driveable or can order on-line	Hardware, tools, wood/lumber, electrical components, paint, plumbing, large appliances, building supplies, adhesives
<i>K-W Surplus</i> 666 Victoria St. N., Kitchener	Driveable	Inexpensive surplus items: electronics, motors, electrical supplies, automotive, hardware, cables
<i>Princess Auto</i> 2 Executive Place, Kitchener	Driveable or can order on-line	Automotive, tools, military
<i>TSC Stores</i> 849 Weber St. N., Waterloo	Driveable or can order on-line	Farming, power equipment, hunting/fishing, automotive, tools, electrical, paint, hardware
<i>Sayal Electronics</i> 1040 Fountain St., Unit 3-4, Cambridge	Driveable	Extensive electronics, motors, robotics
<i>Creatron Inc.</i> Two locations in Toronto	Driveable or can order on-line	Microprocessor boards, small collection of motors, wires, power supplies, wireless
<i>Freescale</i>	On-line only	Microcontrollers, processors, sensors [Includes free samples of many items!]
<i>Digikey Canada</i>	On-line only	Extensive electronics, computer parts, some hardware & tools. Very fast delivery, and only \$8 delivery fee.
<i>RobotShop Canada</i>	On-line only	Robot parts and kits, sensors, motors, microcontrollers, some electronics
<i>Canada Robotix</i>	On-line (or visit near Toronto)	Robot parts, microcontrollers
<i>TigerDirect Canada</i>	On-line only	Computers, computer parts, TV, audio, cameras, GPS
<i>NewEgg Canada</i>	On-line only	Computer hardware, cameras
<i>Newark Electronics Canada</i>	On-line only	Extensive electronics, sensors, test equipment, electrical
<i>AP Circuits</i>	On-line only	Custom PCB creation

Where to Build Your Prototype

The university has facilities that you may find helpful:

- The ECE project room, E2-3353, has benches, oscilloscopes, power supplies, and soldering stations that you may use. Be sure to follow the rules listed on [page 4](#).
- The student design shop in E5-1101D is excellent for drilling, sanding, filing, and painting projects (see [page 4](#)).
- The University's 3D-printer is great for creating custom parts. See <https://uwaterloo.ca/3d-print-centre/> for details.

PCB printing

If you want to create a printed circuit board (PCB), some options are as follows (also, speak with [one of the course instructors](#)):

- AP Circuits (<http://www.apcircuits.com/>). You send them the layout files, and they ship the PCBs within a day or two. Boards are made in Canada.
- OSH Park (<https://oshpark.com/>). You send them the layout files, and they ship the PCBs usually within 2 weeks. Boards are made in the USA. This is cheaper than AP Circuits, but the lag time is longer.

A useful tool for calculating trace widths can be found at <http://circuitcalculator.com/wordpress/2006/01/31/pcb-trace-width-calculator/>.

Other Useful Tips for Prototype Construction

- Soldering: see the tutorial on LEARN.
- Surface-mount soldering: see the tutorial on LEARN.
- Information on electrical wires (types, sizes, etc.): see the tutorial on LEARN.
- Heat-shrink tubing: see the tutorial on LEARN.
- Switches: see the tutorial on LEARN.
- How to properly crimp a battery cable: see the tutorial on LEARN.
- How to select the right fuse: see the tutorial on LEARN.
- Safe charging of Lithium-Ion batteries: see the tutorial on LEARN
- Avoid wiring mistakes that destroy microcontrollers: see the tutorial on LEARN
- Consider using plexiglass (acrylic), wood, and/or aluminum for your project. They are easy to work with and fairly cheap. Plexiglass (acrylic) makes a very nice protective casing for moving parts.
- If compactness isn't a major criterion and you aren't proficient at soldering, consider using breadboards or the old "wire-wrap" method (which is tidier and more reliable than breadboarding).
- Before powering up any electronics, double check the voltage, current, and polarity of connections.
- Remember that if you have any custom 120 V AC circuitry, or any other circuitry (e.g., motor controllers) running at voltages above 40V, or any circuit running at currents greater than 3A, or if you are using lead-acid batteries then you need to talk with one of the course instructors before you even start building your prototype (see item C.3 on [page 6](#)).
- Don't forget to build safety into your prototype right from the start: see items C.5, D.1, and D.2 on [page 6](#).
- You can always ask the one of the course instructors for help or advice.

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REIMBURSEMENT OF PROJECT EXPENSES

The Department of Electrical and Computer Engineering has funds available to partially offset expenses for some types of prototype components and equipment. If you are interested in ECE funding for any prototype expenses, you need to be aware of the following constraints and procedures. Read them carefully!

Constraints and Procedures for Reimbursement

1. Costs for equipment/components/services are eligible for reimbursement this term only if the equipment/components/services are:
 - a. purchased from a reputable supplier that provides an original, itemized receipt
 - b. reasonably priced (e.g., we will not reimburse exorbitant shipping costs)
 - c. purchased between May 1 and the “Last opportunity to complete reimbursement paperwork for prototype expenses” date on [page 8](#).
2. Here is a list of the types of things that can be reimbursed:
 - a. Virtually all hardware (electronics, mechanical parts, computer boards, chips, etc.) are eligible.
 - b. Virtually all software licenses are eligible.
 - c. Virtually all manufacturing costs (e.g., PCB printing, 3D printing) are eligible.
 - d. We will reimburse taxes, shipping/handling fees, custom fees, etc. as long as they are reasonable.
 - e. Personal articles (e.g., clothing, food, travel or parking costs) are not eligible.

In general, the individual (TBA) reviewing the expense has authority to decide whether or not an expense is eligible for reimbursement. Disputes are resolved by the Course Instructor.
3. Here are the procedures and conditions that must be followed to make a reimbursement claim:
 - a. Each group can make at most three reimbursement claims spread over ECE498A and ECE498B, so gather together expenses.
 - b. The maximum total reimbursement for a project group (over both ECE498A and ECE498B taken together) is \$600.
 - c. The minimum claim that will be processed is \$100. (It costs us about \$50 to \$75 to process a claim.)
 - d. All expenses on a given claim must have been paid by the same person. We will not reimburse expenses paid by multiple people. Sorry, no exceptions can be made to this rule.
 - e. Whoever buys the items should use, wherever possible, a credit card. Debit cards or cash purchases are allowed only for small-cost items, and only if the purchaser’s name is included on the receipt.
 - f. Whichever student paid for the items must fill in an *ECE Student Project Funding Approval Spreadsheet* (see [page 36](#)) that lists all the costs. Remember to include taxes, shipping charges, duties, etc.

g. No later than the deadline stated at the bottom of [page 8](#), the same student needs to contact the Lab Instructor to get reimbursed. The details of making a claim are posted in a separate file in LEARN. The student will need to attach scanned copies of the following paperwork:

- i. The *ECE Student Project Funding Approval Spreadsheet*.
- ii. The original itemized receipt(s) or invoice provided by the vendor(s). The receipts must indicate shipping fees, taxes, and any other fees like brokerage fees.
- iii. A credit card statement (or equivalent) showing the charges.
- iv. Packing slips or similar paperwork that came with the purchased equipment/components.

The student then fills in a University *Request for Payment* form (available in LEARN), and uploads a single PDF file (Request for Payment form, Project Funding Approval Spreadsheet, and other paperwork listed in part (g) above) to the individual (TBA) The Lab Instructor will then have the form processed appropriately.

h. Payments are deposited on the student's Quest account within 6-8 weeks. Sometimes it takes longer. If you do not receive a deposit in 12 weeks, contact the individual (TBA) in Finance (currently, Helen Shen, hshen@uwaterloo.ca)

4. Reimbursement claims for this term must be complete by the deadline stated on [page 8](#).
5. If you want to return to a vendor something that you purchased and were reimbursed for, talk to the course coordinator.
6. Claims for expenses paid **more than 4 months** prior will not be processed. Sorry, no exceptions to this rule are allowed.
7. Notwithstanding everything stated above, there is no guarantee that a reimbursement claim will be approved by the Lab Instructor, the Course Instructor, or the University.

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Below is a sample *ECE Student Project Funding Approval Spreadsheet**. You need to fill in all the yellow areas; do not fill in the grey areas. If you will be purchasing many items, add rows to the spreadsheet as needed, **but please do not change the formatting or column headings.**

ECE Student Project Funding Approval Spreadsheet											
Project Title		Autonomous Quadcopter Control System									
Student Name (purchaser)		William Shakespeare									
Project Group Number		2020.075									
Student Signature		<i>William Shakespeare</i>									
Expense Claim Form Tracking Number		(For administrative use only)									
Lab Instructor Approval Signature		(For administrative use only)									
Purchase Description (one receipt per line)	Vendor	Purchase Date	Price	Shipping	Duties	Taxes	Total Cost	Currency	Conversion Rate	Total Cost in CDN\$	
Circuit Components	Digikey	12-Jun-19	\$162.40			\$21.11	\$183.51	CAD	1.00000	\$183.51	
PCB fabrication	PCBway	15-Jun-19	\$60.00	\$10.00	\$10.00	\$7.80	\$87.80	USD	1.23540	\$108.47	
Glue, washers, screws	Home Depot	3-Jul-19	\$3.54			\$0.46	\$4.00	CAD	1.00000	\$4.00	
Total Amount of Claim										\$295.98	

*A blank *ECE Student Project Funding Approval Spreadsheet* is available on LEARN.

CAPSTONE PROJECT AWARDS

There are various awards that are relevant to your project. **Awards are subject to change at any time.**

Awards after the ECE Symposium in ECE498B

It is likely that the following awards will be granted after the symposium in March:

- Keysight Technologies award (Ex.: digital multimeters – one winning group in W'2022)
- Apple award (Ex.: iPad Pro devices – one winning group in W'2022)
- ECE Fourth Year Design Project award (Ex.: \$500 for each winning group – seven winning groups in W'2022)

Note that groups who are awarded other substantial awards may not be eligible for the aforementioned awards.

Entrepreneurship-Focused Awards

Student groups with an entrepreneurial spirit who are designing something that could lead to a business should consider applying for the Entrepreneurship Awards offered at the Faculty of Engineering level. For details, see:

- <https://uwaterloo.ca/engineering/entrepreneurship>
- <https://uwaterloo.ca/engineering/entrepreneurship/funding>

Deadlines and application details are on the web pages. **Note that some application deadlines are in the 4A term.**

Entrepreneurship awards are not directly related to ECE498A/B: the teaching team of ECE498A/B has no input on how awards are given out, and groups that receive entrepreneurship awards are subject to the same course requirements as all other groups.

Other Capstone Project Awards

Occasionally, companies sponsor awards that focus on specific topics. Unfortunately, we tend not to hear about such awards until the 4B term, but you will be informed of any similar awards as soon as possible. The Faculty of Engineering is continually working with sponsors to create awards. The following site gives additional information on awards.

- <https://uwaterloo.ca/capstone-design/supporters-and-financial-awards>

SUMMARY OF RESOURCES ON LEARN

It is important that you regularly look at the course LEARN pages. In addition to announcements, calendar entries, and the Safety Quiz, the following forms and resources are provided:

- Course logistics:
 - The Course Outline (this document)
- Important safety information:
 - The *Safety Rules for E2-3353* document
 - Other relevant safety information
- Tutorials for soldering, heat shrinktubing, switches, wire selection, fuse selection, and other useful topics
- Slides from presentations given at the Tuesday sessions (as they become available)
- List of project ideas and potential consultants
 - Project Abstracts from previous years
 - Ideas suggested by faculty members (if available)
 - A partial list of potential consultants
- List of projects and group numbers (once groups have been formed and group numbers assigned)
- Template for Student Log
- Blank *ECE Student Project Funding Approval Spreadsheet*
- Project Abstract and Project Agreement Form information:
 - A blank *Project Agreement Form*
 - Sample Project Abstracts from previous years
- Project Specifications and Risk Assessment Document information
 - Sample submissions
- Detailed Design and Project Timeline Document information
 - Sample rubric spreadsheet to see exactly how the rubric will be used
 - Sample submissions
- Initial Prototype Demonstration and Progress Report information:
 - A Blank *Initial Prototype Demonstration Feedback Sheet*
- Revised Project Title and Project Abstract information:
 - LEARN Dropbox for on-line submission
 - Sample submission

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STANDARD STATEMENT OF UNIVERSITY POLICIES

The following statements represent university expectations and policies with respect to academic integrity, discipline, grievance, student appeals, and academic accommodations. If you would like more clarification, please contact your course instructor directly.

Academic Integrity

In order to maintain a culture of academic integrity, members of the University of Waterloo community are expected to promote honesty, trust, fairness, respect and responsibility. [Check www.uwaterloo.ca/academic-integrity/ for more information.]

Grievance

A student who believes that a decision affecting some aspect of his/her university life has been unfair or unreasonable may have grounds for initiating a grievance. Read Policy 70, Student Petitions and Grievances, Section 4, <https://uwaterloo.ca/secretariat/policies-procedures-guidelines/policy-70>. When in doubt please be certain to contact the department's administrative assistant who will provide further assistance.

Discipline

A student is expected to know what constitutes academic integrity [check www.uwaterloo.ca/academic-integrity/] to avoid committing an academic offence, and to take responsibility for his/her actions. A student who is unsure whether an action constitutes an offence, or who needs help in learning how to avoid offences (e.g., plagiarism, cheating) or about "rules" for group work/collaboration should seek guidance from the course instructor, academic advisor, or the undergraduate Associate Dean. For information on categories of offences and types of penalties, students should refer to Policy 71, Student Discipline, <https://uwaterloo.ca/secretariat/policies-procedures-guidelines/policy-71>. For typical penalties check Guidelines for the Assessment of Penalties, <https://uwaterloo.ca/secretariat/guidelines/guidelines-assessment-penalties>.

Appeals

A decision made or penalty imposed under Policy 70 (Student Petitions and Grievances) (other than a petition) or Policy 71 (Student Discipline) may be appealed if there is a ground. A student who believes he/she has a ground for an appeal should refer to Policy 72 (Student Appeals) <https://uwaterloo.ca/secretariat/policies-procedures-guidelines/policy-72>.

Note for Students with Disabilities

AccessAbility Services, formally known as the Office for Persons with Disabilities (OPD), is located in Needles Hall extension, room NH-1401. This office collaborates with all academic departments to arrange appropriate accommodations for students with disabilities without compromising the academic integrity of the curriculum. If you require academic accommodations to lessen the impact of your disability, please register with AccessAbility Services at the beginning of each academic term: <https://uwaterloo.ca/accessability-services/>.